

ORIGAMI AS A SPATIAL REASONING BENCHMARK

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ABSTRACT

Progress on reasoning has been fastest where verification is free. In mathematics and in code, a candidate answer can be checked exactly, cheaply, and without a human, and that property has shaped how quickly those domains improved. Spatial and physical reasoning has no such checker, so it is evaluated instead with static images, multiple choice questions, or learned surrogate simulators whose wrong verdicts look exactly like their right ones. A benchmark can only be as trustworthy as the judge inside it. We show that origami is a domain where exact verification is free, and that the gap between generating a problem and solving it is severe. Folding a sheet forward is trivial: fold, record, unfold. Recovering the discrete sequence of folds that produced a crease pattern is NP hard. The same forward engine that generates a sample can therefore rule on any proposed step exactly, at no cost, with no learned model in the loop. The task is also not a puzzle chosen for convenience. It becomes strictly harder as it proceeds, because every fold thickens the stack and constrains the next one, and a solver has to track a planar geometry together with a layer ordering that continuously constrain each other. We release CP2Seq, a benchmark whose ground truth is constructed rather than annotated, together with an environment that is also the verifier: stepping it is the same act as asking for a verdict. It executes one candidate fold and either returns the resulting state or refuses with a named reason, such as the sheet would tear, the fold creases nothing, or the selected layers cannot move in that direction. It performs no search and makes no choices of its own, so the work of proposing, pruning and backtracking stays with the model, and what the benchmark scores is search rather than geometry. Samples are generated under two action models taken from the simple folding literature and stratified along two axes, fold depth and coupling, where coupling is the number of layers a single fold cuts. Every sample rebuilds byte identically from its seed and is checked by tolerance free replay. Because one crease pattern admits many valid folded states, and because the recorded sequence is not guaranteed to be minimal, an answer is scored by replaying the proposed sequence rather than by step wise agreement with the stored one. The scoring protocol is solve rate at a fixed query budget, reported per difficulty stratum under two notions of equality, equality of crease sets and equality of folded states up to a symmetry group declared in advance, with queries to solution as a secondary measure over all attempts including those that exhaust the budget. We instantiate this protocol on off the shelf multimodal language models, with ablations that remove visual feedback, remove filtering, and remove tools entirely as a memorization control on anonymized geometry.

1 POSITIONING

This paper is submitted as a **benchmark and dataset** contribution. What it offers the community is an evaluation substrate for **spatial and geometric reasoning in large language models**: a corpus, an environment that doubles as an exact verifier, and a scoring protocol, rather than a model, a training method or a state-of-the-art number. Every arm reported here runs an off-the-shelf model through the harness; nothing is trained. The claim under review is that the benchmark measures what it says it measures and that its judge can be trusted, not that any particular model is good at the task.

The reasoning being measured is geometric as much as it is spatial, and the distinction is load-bearing rather than cosmetic. A solver has to reason about lines, reflections, incidence and adja-

gency on the original sheet, which is geometry in the ordinary sense, and simultaneously about a layer ordering that the geometry does not determine, which is combinatorial. Calling the task spatial reasoning alone understates the second half. Positioning the work as a benchmark for spatial *and geometric* reasoning states the scope accurately and places it beside the multimodal-reasoning benchmarks of Section 3.1 rather than beside the computational-origami literature of Section 3.3, which is the theory the benchmark rests on rather than the field it contributes to.

2 INTRODUCTION

Reasoning has improved fastest in the two domains where an answer can be checked without a human. A proof can be run through a proof assistant; a program can be run against its tests. In both cases the check is exact, it is cheap enough to run millions of times, and it does not itself have to be trusted, because it is not a model of the thing, it is the thing. That property is what makes large scale evaluation, rejection sampling, search at test time and reward from verification possible at all.

Spatial and physical reasoning has no such checker, and the field has adapted in three ways, each of which costs something specific. Benchmarks built on static images or multiple choice questions never execute the model’s proposal, so an answer that is right for the wrong reason is indistinguishable from one that is right. Benchmarks and methods built on learned surrogate simulators do execute the proposal, but against an approximation: a wrong verdict looks exactly like a right one, which means the benchmark’s own error rate is unknown and unmeasurable from inside. Human judgement and preference scores are exact in a sense, but they do not scale and they are not reproducible across papers. The common consequence is worth stating plainly, because it is the motivation for everything below: **a benchmark can only be as trustworthy as the judge inside it.**

We observe that origami has the property the field is missing, and that it has it in an unusually clean form. Folding a sheet forward is trivial: pick a line, fold across it, record what you did, and unfold at the end to read off the crease pattern. Recovering the discrete sequence of folds that produced a given crease pattern is NP hard (Arkin et al., 2004; Akitaya et al., 2017). The forward direction is not merely easier than the inverse; it is the *generator*, and the same engine that generates a problem can rule on any proposed step of a solution exactly. Verification costs one forward application of an operation the engine already implements. There is no learned component anywhere in the loop, no tolerance to tune, and no oracle to trust.

This asymmetry is what a benchmark wants. Difficulty is controllable rather than sampled and hoped for, because the generator chooses how deep and how tangled a sample is. Ground truth is free rather than annotated, because the sequence is recorded as it happens rather than inferred afterwards. Negative examples are unlimited, because any illegal fold is generated by asking for one. And the judge is the same object as the environment, so the benchmark’s verdicts do not depend on a second artifact that could disagree with the first.

The task is also hard in ways that a maze or a grid-world is not, and the differences are measurable rather than rhetorical. First, it gets harder as it proceeds: every fold thickens the stack and constrains what the next fold may do, so the difficulty of step k is a function of what happened in steps 1 through $k - 1$. Second, a solver has to maintain two kinds of state that constrain each other. The crease graph is fixed once the pattern is given, but the vertex coordinates change at every step, the overlap structure changes at every step, and the layer ordering changes at every step and is *chosen* rather than computed. Determining that ordering is NP hard even when a valid mountain-valley assignment is supplied (Bern & Hayes, 1996). Mazes have geometry alone; here an independent combinatorial layer has to be tracked alongside it. Third, the action space is not something we invented for the occasion: simple folds have been defined and their complexity studied for two decades (Arkin et al., 2004), so the operations an agent is given have names, and claims about them can be checked against a literature.

We contribute the following.

1. **CP2Seq**, a benchmark for crease-pattern-to-sequence recovery whose ground truth is constructed rather than annotated, generated under two named action models from the simple folding literature, stratified by fold depth and by coupling, and reproducible from seeds rather than shipped as data.

2. **An environment that is also the verifier.** Stepping it is the same act as asking for a verdict. It is exact, it is free, it performs no search, and its refusals are named rather than boolean, so a rejection is a diagnosis instead of a dead end.
3. **A scoring protocol that survives non-unique solutions.** Answers are judged by replay, at two levels of equality, with the symmetry group fixed in advance, because the recorded sequence is one solution among several and is not guaranteed to be the shortest.
4. **An evaluation of off-the-shelf multimodal language models** in this loop, with a visual-feedback ablation, a filtering ablation, and a memorization control on anonymized geometry. [TO RUN]
5. **A measured scope boundary.** 89.3% of real crease patterns provably lie outside all-layers simple folding. The benchmark’s action space is bounded by evidence rather than by assertion, and the boundary is reported rather than buried.

3 RELATED WORK

3.1 ORIGAMI BENCHMARKS FOR MULTIMODAL MODELS

OrigamiSpace (Xu et al., 2025) contributes 350 instances, each carrying a crease pattern diagram, a compiled flat pattern, the complete folding process and a final folded image, and defines four tasks over them: pattern prediction, multi-step spatial reasoning, spatial relationship prediction, and end-to-end crease-pattern code generation. It also provides an interactive environment and describes reinforcement learning as a possibility explored rather than a result demonstrated. Its design lesson is that one richly instrumented sample can support several tasks, which is how 350 instances sustain a benchmark paper.

GamiBench (Spencer et al., 2025) contributes 186 regular and 186 *impossible* crease patterns paired with folded shapes from six viewpoints, across three visual question-answering tasks, and introduces two diagnostic metrics: viewpoint consistency and impossible fold selection rate. Its impossible patterns are the closest the current literature comes to forcing a model to judge feasibility rather than recognise a shape, and its headline finding is that leading models struggle even at single-step spatial understanding. It is the benchmark this work is most directly in conversation with, and the one whose negative examples we can produce without limit, because generating an illegal fold in our environment is the same operation as generating a legal one.

OrigamiBench (ori, 2026), **FoldingAgent** (Moriya et al., 2026) and **COrigami** (cor, 2026) complete the picture; their metrics appear in the comparison of Section 3.4.

What all of these share is the gap. In each, the model’s proposal is answered by a question rather than by an execution: it selects among rendered alternatives, or produces an artifact that is compared to a target. None of them steps a paper-exact engine and asks whether *this* move, on *this* stack, is something paper could do.

3.2 METHODS THAT PAIR A PROPOSER WITH A SIMULATOR

Learn2Fold (Huang et al., 2026) formulates origami as conditional program induction over a crease-pattern graph. A language model generates candidate folding programs from text, and a learned graph-structured world model acts as a differentiable surrogate simulator that predicts physical feasibility and failure modes before execution, inside a lookahead planning loop. Its stated key insight is to decouple semantic proposal from physical verification.

We adopt that decomposition and differ on one thing: the verifier. Theirs is learned, differentiable and approximate, which is what makes it usable for planning and also what makes its mistakes invisible. Ours is exact and costs nothing, because it is the generator run forwards. Their scoring is step-level precision, recall and F1 against an expert sequence, together with edge-level IoU; Section 7.1 shows why that family of measures is unsafe in this task, since a model that finds a *shorter* correct sequence is marked wrong by it. That is a finding about scoring in this setting rather than a criticism of their results.

The general precedent for a proposer paired with a verifier is older than any of this work: a network that proposes and a search that checks is the structure behind AlphaGo, and the reason the combination is stronger than either half.

3.3 COMPUTATIONAL ORIGAMI: THE THEORY UNDER THE ACTION SPACE

The benchmark’s central claim, that geometry does not determine a folded state, is Bern and Hayes’s: deciding the overlap order of a flat folding is NP hard even given a valid mountain-valley assignment (Bern & Hayes, 1996). The action space is Arkin et al.’s: simple folds, in the one-layer, some-layers and all-layers variants, with finite or infinite fold lines (Arkin et al., 2004). Their complexity has been mapped since, including simple foldability’s hardness (Akitaya et al., 2017) and the infinite all-layers case (Akitaya et al., 2019). The solver whose constraint vocabulary we borrow for terminal states is Flat-Folder (Akitaya et al., 2024).

Two results shape what the task actually is. Demaine et al. (2004) showed that any well-behaved folded state is reachable by *some* continuous motion; the question is therefore never whether a path exists, but whether one exists through a finite number of the discrete operations a hand or a machine can perform. And counting foldings has no closed form even for a one-dimensional strip, a problem open since Lucas attributed it to Lemoine in 1891 (Lucas, 1891; oei); this is the answer to “why not enumerate the states”, and it is a fact about the problem rather than about any implementation.

Finally, the problem itself is not new: generating folding sequences from crease patterns was named and attacked symbolically by Akitaya et al. (2013), by the same group whose tools this field now depends on. What is new is neither the problem nor the theory, but the observation that this problem supplies an exact verifier for free and can therefore serve as an evaluation substrate for models that reason with tools.

3.4 THE STRUCTURAL GAP

Collecting the metrics used across this literature makes one absence visible. Every metric in use compares a produced artifact against a target: geometric or semantic similarity to a reference, precision and recall against an expert sequence, IoU over affected edges, compilation success, human preference. **None ranks among the several valid folded states of a single crease pattern, and none scores an agent by how much verification it required.** The absence is structural rather than accidental, because computing any of those metrics requires a target to compare against, and the interesting question here has more than one right answer.

4 PROBLEM FORMULATION

4.1 FLAT-FOLDED STATES

A flat-folded state is an isometric map of the sheet into the plane together with a layer ordering. The map is fixed by the crease pattern; the ordering is not. A single crease pattern therefore admits many folded states that are geometrically identical and distinct as embeddings, and determining the ordering is NP hard even when a valid mountain-valley assignment is given (Bern & Hayes, 1996).

This distinction is the reason the task cannot be reduced to geometry. Over a folding sequence the crease graph’s topology never changes: the same vertices, edges and faces exist at step one and at step nineteen. What changes is everything else. Vertex coordinates are reflected at every step. The overlap structure, which faces lie above which, changes at every step and follows from the geometry. The layer ordering changes at every step and does *not* follow from the geometry; it is chosen, and it is where the decisions live. The number of layers accumulates, which is the formal content of “it gets harder as it goes”.

4.2 THE ACTION SPACE

Simple folds come in named variants, and any benchmark in this area has to say which it uses, because the complexity results differ between them.

Table 1: Named variants of the simple fold (Arkin et al., 2004).

MODEL	DEFINITION
One-layer simple fold	Rotate a single layer $\pm 180^\circ$ about a line
Some-layers simple fold	Rotate a contiguous run of layers
All-layers simple fold	Rotate every layer the line crosses, as sheet metal bends
Finite vs infinite line	Whether the fold line is a full line or a segment

CP2Seq instantiates the all-layers and some-layers tiers. The all-layers tier is the simpler object and it buys three properties at once, the third of which is rarely named. Because the whole stack moves as a rigid body, no two connected pieces of paper ever move relative to each other except at the fold line itself, where paper is allowed to bend. Self-intersection is impossible, the layer ordering is forced rather than chosen, and **tearing is impossible by construction**.

Moving only some layers brings all three back, and tearing becomes the binding constraint. If a moving face is joined to a stationary face along a crease, and that crease is not on the fold line, the fold would rip the sheet. This is decidable only in the coordinates of the original square, because adjacency is a fact about the original sheet that survives any amount of folding and cannot be recovered from the current positions of the polygons alone.

An example makes the tier difference concrete, and it needs only two folds. Fold the right half of a square over to the left across the vertical midline. The result is a two-layer stack whose top layer is joined to the bottom layer along the entire crease at the midline. Now select the top layer alone. Folding its upper half down across the horizontal midline would tear the sheet, because the moving piece is joined to stationary paper along a vertical crease segment that is perpendicular to the fold line. Folding its left quarter across a vertical line does not tear, because the moving piece’s only join to stationary paper lies on the fold line itself. Same state, same layer selection, opposite verdicts, and the only difference is the orientation of the line.

4.3 THE TASK

Given a crease pattern, produce a sequence of simple folds that reproduces it. Generating an instance costs one forward pass of the engine; solving it is NP hard. That asymmetry is the paper’s foundation.

Caveat. One claim must not be made, and it was drafted wrongly here once. It is tempting to say that legal moves are rare and that this is what makes the task hard. Enumeration refutes it: legal partial folds *grow* with depth, from 22 at two layers to 332 at thirty-eight, six times the 56 all-layers folds available at the same state. What falls is the hit rate of uniform random proposal, from 11.9% to 3.8% by seventeen layers, because the space being sampled grows faster than the legal set inside it. That is a fact about a sampler, not about origami, and conflating the two would put a false statement about branching factor into the paper.

5 CP2SEQ

5.1 GENERATION

The corpus is synthesised and the project takes no outside data. A sample is produced by folding forward from a square with randomly chosen simple folds, recording each fold as it is applied, and unfolding at the end to obtain the crease pattern. The (pattern, sequence) pair is built rather than labelled, so there is nothing to annotate and nothing to trust, and every sample is correct by construction: the pattern is what the folding left behind.

Nothing is shipped that cannot be rebuilt. Given its seed, a sample regenerates byte-identically, so the corpus is distributed as a generator plus a manifest rather than as data.

5.2 DIFFICULTY, STRATIFIED RATHER THAN SAMPLED

Difficulty is set by the sampler along two axes rather than measured after the fact. **Fold depth** is the primary axis. **Coupling**, the number of creases a single fold creates, which is to say how many layers it cuts, is the second; it is free to compute at generation time, since one fold cuts every layer it crosses and each cut becomes one crease in the unfolded square. Coupling is simultaneously the non-local dependency that Learn2Fold’s difficulty tiers appeal to and the closest measurable proxy for how hard a model is to fold by hand.

The release corpus holds 600 samples in five batches: 400 all-layers samples spanning 4 to 19 folds, stratified easy, mid and hard on the action space’s tertiles, and 200 some-layers samples at depths 3 to 6. Median crease count is 33 and median layer count 24, with the deep tail reaching tens of thousands of both.

Caveat. Two limits belong in the paper rather than in an appendix. The first is an empty cell: long sequences at low coupling are almost unreachable, yielding 2hits in 16,000 attempts, because every all-layers fold thickens the stack, so a long sequence cannot keep cutting few layers. Real folders reach that corner by **pre-creasing**, an operation this action space excludes. The longer a real model runs, the more of it sits outside what the benchmark can express. The second is depth: the some-layers tier stops at six folds in this release, and the difficulty the benchmark is about lives past the depth a search can reach. Restoring the deeper tier is a matter of generator configuration and wall-clock, not redesign.

5.3 WHAT A SAMPLE CARRIES

Table 2: The files shipped with each CP2Seq sample.

FILE	CONTENTS
<code>cp.fold</code>	The crease pattern, planarised. The input
<code>seq.json</code>	Every fold: line, direction, creases made, coupling. The ground truth
<code>steps.fold</code>	The pattern plus the folded state after each step, as one multi-frame file
<code>meta.json</code>	Difficulty metrics, degeneracy flags, provenance

5.4 VERIFYING THE CORPUS ITSELF

Two checks run over every sample, written independently of each other. The first replays each recorded sequence and confirms it reproduces the pattern stored beside it. The second, written without reference to the first, compares maximal creased intervals as multisets with no tolerance in the verdict, after clustering edges into lines so that subdivision differences cannot register as disagreements. Both pass on all 600 samples of the release corpus.

Caveat. Both share the fold engine with the generator, which bounds what they can establish. They cannot catch a wrong model of paper: if the engine is wrong about tearing, replay is wrong in the same way and agrees with itself. What they do catch is drift between what was folded and what was recorded, which is the likelier failure and the one that silently corrupts ground truth. An independent check requires a third-party solver and is not claimed here.

6 THE ENVIRONMENT, WHICH IS ALSO THE VERIFIER

6.1 ONE OBJECT, TWO NAMES

In this benchmark the environment and the verifier are the same object. Stepping it is the same act as asking for a verdict: the model submits a candidate fold, and the reply is either the resulting state or a refusal. There is no separate validity oracle, no second artifact that could disagree with the first, and no learned component in the loop.

Table 3: Who does what. The environment does not search.

COMPONENT	RESPONSIBILITY
Environment / verifier Model	State update and legality. It does not search and does not choose Proposes the next fold, decides where to go, recognises when to backtrack

6.2 THE DIVISION OF LABOUR

This division is what makes the benchmark measure something. The obvious objection to any tool-augmented setup is that the tool is doing the work, and here the answer is structural rather than rhetorical: deciding global flat-foldability is NP hard, so the environment cannot search its way to an answer even in principle. It can only say whether one proposed step is something paper could do. Compressing an exponential search into a feasible number of queries is exactly what the environment cannot supply and exactly what the model is being measured on. The referee is not a player.

6.3 REFUSALS ARE NAMED

An illegal fold returns a structured refusal with a name, not a boolean.

Table 4: The refusal taxonomy. A rejection arrives already labelled with the class it violated.

REFUSAL	MEANING
would-tear	The moving layers are joined to stationary paper away from the fold line, so the sheet would come apart. The binding constraint of the some-layers tier
no-crease	The fold moves layers without creasing anything, tearing them free of the sheet rather than folding them
nothing-to-move	The fold line misses the selected layers entirely
direction-impossible	A run taken from the bottom of the stack was asked to fold over, or the reverse
bad-line	The fold line is malformed, or names no side of itself

The names are the design. A model told `would-tear` can act on the information; a model told “illegal” cannot. This also turns error analysis into a taxonomy that exists before the experiment: every rejection arrives already labelled with the class it violated, so a reject is a diagnosis rather than a dead end.

Caveat. There is deliberately no self-intersection refusal, and there cannot be one in this action space. A some-layers fold may only move a contiguous run of layers taken from the top or the bottom of the stack, and such a move cannot drive paper through the layers it left behind. Lifting the top two layers and folding them across is something a hand can do; folding them *underneath* the stack is not a fold but a slit. Restricting selection to a run at one extremity makes the illegal case unrepresentable rather than undetected. The four constraint classes that Flat-Folder checks (taco-taco, taco-tortilla, tortilla-tortilla, transitivity) (Akitaya et al., 2024; fla) remain the right vocabulary for global terminal states; they are not what a single legal step needs checking against here.

6.4 WHAT THE MODEL OBSERVES

On success the environment returns the new state together with two views: a top-down X-ray of the stack, and an exploded view of the layers. Every intermediate state in this tier is flat, so a second camera angle would show the same silhouette rotated and would carry no new information; what carries information is layer structure, which is why the second view is an explosion rather than a rotation. Positions are reported as coordinates on the original sheet together with an integer

layer index. There is no continuous vertical coordinate at this tier, and a rendering that looks three-dimensional is not evidence of one; thickness belongs to a different problem and a different paper.

6.5 ERROR VERBOSITY IS AN EXPERIMENTAL VARIABLE

The richer a refusal, the more of the reasoning the environment performs rather than the model. At the limit, a message that names the fix has solved the step. Verbosity is therefore fixed before the first run and reported with the results; otherwise the tool-augmented condition is not reproducible and its comparison against the verifier-only condition measures message design rather than reasoning.

7 SCORING

7.1 WHY THE OBVIOUS PROTOCOL IS WRONG

The recorded sequence is *a* solution, not *the* solution. The generator folds; it does not search, so nothing makes its sequence minimal, and no minimal-length label is shipped because producing one would require a search this benchmark deliberately does not run. Measured on the verified split, 2 of 100 samples admit a sequence one fold shorter than the one that built them.

The consequence is sharp. **Any metric that compares a proposal step by step against the recorded sequence marks a better answer wrong.** A model that finds a shorter correct solution is penalised for it. This disqualifies edit distance, and it disqualifies step-level precision, recall and F1 against an expert sequence for this task. Step count is not a correctness signal in either direction.

7.2 WHAT WE DO INSTEAD

Proposals are scored by replaying them through the engine and comparing the resulting states. Both sides are replayed: the model’s sequence and the reference sequence both pass through the same engine, and a stored frame is never the object compared against. That last point is a consequence of the file format rather than a preference. Stored frames record where each face sits in the current plane and not which piece of the original sheet it is, and deep in a stack many layers are congruent triangles, so two states differing by a swap of two such layers cannot be distinguished from stored frames alone. A replayed state carries original-sheet coordinates for free, because the engine needs them to decide tearing at all.

Two levels of equality are reported.

Level 1, crease sets. An exact multiset comparison of maximal creased intervals, equal rather than overlapping, with no tolerance in the verdict.

Level 2, folded states up to a declared group. Because one crease pattern admits many valid terminal states, exact equality against the one stored state would mark correct answers wrong. The answer is not a tolerance but an equivalence: declare in advance which differences do not count, then demand exactness inside that. The group is the eight symmetries of the square together with an arbitrary translation. Translation is included because a fold can carry paper off the original square, so a folded state’s position in the plane is an accident of the sequence. The four reflections additionally reverse the stack and flip every face’s parity, because turning a model over does all three at once; applying a reflection to coordinates alone would compare a model against its mirror image and silently accept wrong answers.

Caveat. The layer-order variants that a flat-foldability solver reports as equally valid are deliberately *not* in the group. Those are not symmetries of one state; they are different states reachable from the same crease pattern, and deciding which orderings are valid requires a solver. Folding that into an equality test would hide an external dependency inside what reads as arithmetic. If that equivalence is wanted, it belongs in a separate check that owns the solver.

7.3 METRICS

The primary metric is **solve rate at a fixed query budget**, reported per difficulty stratum and never pooled, under both levels of equality. A proposal is solved if replaying it reproduces the target.

Exhausting the budget is a timeout, which is reported separately from a wrong answer because the two mean different things.

The secondary metric is **queries to solution**, reported over all attempts including those that time out. Conditioning on the solved attempts hides precisely the long-tail blowup that makes the task interesting. This metric is what answers the “the simulator did all the work” objection quantitatively: two models sharing one exact verifier can be ranked by how little of it they needed.

7.4 VALIDATING THE PROTOCOL

A verifier paper that does not test its own verifier is asking to be taken on trust, so the comparator ships with a negative control: a deliberately corrupted state, with two layers swapped, must be rejected.

The control earned its place. On its first full run, 23 of 600 corrupted states were **accepted**. The cause was not a tolerance. The comparison used current-plane geometry and parity alone, and deep in a stack many layers are congruent triangles, so two layers occupying the same region but originating from different parts of the sheet were interchangeable. They are not interchangeable: which one lies underneath is a real difference between two folded states, and precisely the difference Level 2 exists to catch. With face identity on the original sheet restored to the comparison, the full corpus reports 600 states equal, all under the identity symmetry, and 600 of 600 corrupted states rejected. Every verdict additionally reports whether face identity was available, so a weakened comparison is visible in the output rather than assumed away.

8 OBJECTIONS, ANSWERED HERE RATHER THAN IN REBUTTAL

“The simulator does all the work.” It cannot: deciding flat-foldability is NP hard, and the environment only rules on single steps. Section 6.2 states the division of labour and Section 7.3 measures it.

“Origami is a toy problem.” Flat origami is Turing complete (tur, 2023), so the domain is not expressively impoverished. **Caveat.** This is a defensive citation and nothing more. It shows that the domain is rich enough to be interesting; it does not show that anything learned here transfers, and the causal chain from “origami is Turing complete” to “training on origami yields physical understanding” is broken in the middle. Rule 110 is Turing complete too.

“Why not enumerate the valid states?” There is no closed form for the number of foldings even of a one-dimensional strip, a problem open since 1891 (Lucas, 1891; oei; Koehler, 1968), and state counts for real patterns reach astronomical magnitudes. The hardness is the problem’s, not an implementation’s.

“Synthetic data is a shortcut.” The comparable published corpora are largely produced by their authors’ own symbolic simulators; synthesis is the field’s normal practice. Here it is also what makes difficulty controllable and ground truth free, and what allows the corpus to ship as a seed rather than as a file.

“Your corpus is not real origami.” Correct, and measured: 89.3% of real crease patterns provably lie outside all-layers simple folding. This is stated as a scope boundary in Section 2 and Section 12 rather than left for a reviewer to discover.

“The model may have memorised the pattern.” The no-tools arm runs on anonymized geometry with filenames and metadata stripped, for exactly this reason: if a model can read a pattern’s name it recalls rather than reasons, and then both arms measure recall and the comparison measures nothing. See Section 10.

9 EXPERIMENTS

[TO RUN]

- **Models.** Held fixed across arms; comparing across models would confound the ablation.
[TO RUN: the main evaluation sweep]

- **Protocol.** Query budget, repeats, seeds; median and spread reported, never a single run, because sampling is not deterministic and one run is not a measurement. [TO RUN: same sweep]
- **Main table.** Solve rate per stratum under Level 1 and Level 2. [TO RUN: same sweep]
- **Query distribution.** Queries to solution over all attempts, with the timeout fraction. [TO RUN: same sweep]

10 ABLATIONS

[TO RUN]

Table 5: The ablation arms. Same model, same patterns, same query budget across arms.

ARM	TOOLS AVAILABLE	WHAT IT MEASURES
Full tool belt	Environment, views, any auxiliary tools	Upper bound
No vision	Same, rendered views withheld	The value of visual feedback specifically
Verifier only	Pass/fail, no filtering	The value of filtering on top of raw verification
No tools	None, anonymized geometry	Memorization control

Same model, same patterns, same query budget across arms. If the no-vision or no-tools arm performs nearly as well as the full arm, **that is a finding rather than a failed experiment**: it means the gain is not where it was assumed to be. [TO RUN]

11 RESULTS

[TO RUN]

- **Headline.** [TO RUN]
- **Failure taxonomy.** Rejections grouped by refusal class; the taxonomy already exists, so this is tabulation rather than interpretation. [TO RUN]
- **Where difficulty lives.** Depth against coupling. [TO RUN]

12 LIMITATIONS

No training. Every arm is an off-the-shelf model driven by prompting and tool calling; no model is trained or fine-tuned. The contribution is the harness, not a model. This is stated in the introduction as well as here, because a limitation the authors declare is context and one a reviewer discovers is an objection.

Scope of the action space. 89.3% of real crease patterns lie outside all-layers simple folding, and pre-creasing, the operation real folders use to reach long sequences at low coupling, is excluded.

The corpus verifiers share an engine with the generator, so they cannot detect a wrong model of paper. They detect drift between what was folded and what was recorded.

Flat states only. No thickness, no material, no mechanics. Thickness changes the geometry outright and belongs to a separate line of work with a verifier several orders of magnitude more expensive.

Level 2’s group is narrower than the truest notion of equality, because valid layer-order variants are excluded for the reason given in Section 7.2. It is therefore strict rather than permissive, which is the safe direction but not a free one.

13 FIGURES

Table 6: Planned figures and their status.

#	FIGURE	
1	The asymmetry: generation is one forward pass, recovery is NP hard	To draw
2	The tearing pair of Section 4.2. One state, one layer selection, a vertical fold line legal and a horizontal one not	To draw
3	A refusal as the model receives it: view, named reason, per-constraint checklist with locations	To draw
4	The difficulty grid, depth against coupling, with the empty cell visible	To draw
5	Main result per stratum	[TO BE DRA]

14 REPRODUCIBILITY

The corpus ships as a generator and a manifest; every sample rebuilds byte-identically from its seed. The environment, both corpus verifiers and the state comparator are released with the benchmark. The negative control of Section 7.4 runs as a command, so a reader can confirm that the comparator rejects corrupted stacks rather than taking Section 7.4 on trust.

ACKNOWLEDGMENTS

Withheld for anonymous review.

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A REFERENCE NOTES

The bibliography is grouped in `iclr2026_conference.bib` by the role each work plays in the argument rather than alphabetically, so that a co-author can see what is load-bearing. The notes below record what is still outstanding.

Origami benchmarks and systems for language models. Xu et al. (2025) must be cited as a preprint; `papers.md` recorded NeurIPS'25 and the arXiv record carries no venue. Spencer et al. (2025) has code and data released. Moriya et al. (2026) reconstructs folding processes from instructional video and is built on the PurelandFold dataset; code at <https://github.com/maya-moriya/FoldingAgent>. **OrigamiBench** (ori, 2026) has a *citation missing*: it appears in our metrics survey as the source of query efficiency and geometric/semantic similarity, and its dataset is the 366 patterns in Flat-Folder's `examples/instagram/`. A full reference has to be recovered before submission. **COrigami** (cor, 2026) has a *citation missing* as well: it appears in the metrics survey as the source of flat-foldability as a boolean and a VLM aesthetic score. Same action needed.

Complexity and theory of flat folding. Bern & Hayes (1996) is the citation under the paper's central claim: geometry does not determine layer order, and deciding overlap order is NP hard even given a valid mountain-valley assignment. Arkin et al. (2004) defines simple foldability; the action space comes from here. Akitaya et al. (2019) covers the all-layers model, which corresponds to sheet-metal bending. Akitaya et al. (2024) is the Flat-Folder paper; deciding a global flat-folded state is NP hard. See also mix (2023), lay (2026) and fla (2018). tur (2023) is a defensive citation only, per Section 8. Demaine et al. (2004) separates folded state from folding motion: reachability is free, so the hard question is the discrete step structure; its predecessor is Demaine & Mitchell (2001).

The crease-pattern-to-sequence problem itself. Akitaya et al. (2013) named the problem, by the group whose tools the field now uses: reflection paths, graph rewriting, step graphs; a frog base explodes to 22,665 nodes and 30 minutes, and the authors' own future work asks for the priority heuristic this paper's models are being asked to supply. cre (2022) is an open-source implementation of the above and is usable as a symbolic baseline. [TO RUN if used]

Counting folded states. oei lists the foldings of a strip of stamps: 1, 2, 6, 16, 50, 144, 462, 1392, 4536, 14060, ... There is no closed form. See also A001011 and A001416, and Lucas (1891); Koehler (1968); Lunnon (1971); mea (2013); meanders and stamp foldings are the same combinatorial object.

Spatial reasoning context. spa (2025) surveys the area. vot (2024) names the mental-imagery question this benchmark can turn into a measurement.

Tools, formats and data. fla decides and enumerates flat-folded states and is the source of the four constraint classes cited in Section 6.3. fol is the de facto standard used by every artifact we ship. Mitani is a crease-pattern editor with folded-form estimation. pur is not used as training or evaluation data here; it is cited as the closest existing sequence-level corpus and as FoldingAgent's data.

Statistical physics of random flat-foldability. Optional support for the claim that multiple valid states is the normal case rather than a curiosity, to be included only if Section 4.1 needs reinforcement: spi (2024) and ran (2025).

Deliberately not cited. Thick folding (Ku & Demaine 2016), bar-and-hinge mechanics, MERLIN2, SWOMPS, Sim-FAST-PY, and the creased-sheet mechanics literature all belong to the thickness and mechanics line of work. They are a separate paper with a verifier several orders of magnitude more expensive, and citing them here would invite a reviewer to ask why this paper does not do that one.

B EDITORIAL RULES FOR THIS DRAFT

Not part of the paper; carried over from `paper/DRAFT.md` so that nothing is lost in the conversion. Each of these has been drafted wrongly at least once in this project.

1. **Never claim legal moves are rare.** The enumeration refutes it, per Section 4.3.

2. **Never claim the model performs 3D reconstruction.** Say spatial reasoning; pose “is a three-dimensional representation necessary or merely sufficient” as a research question. Avoid homotopy and isotopy language entirely.
3. **Turing completeness is defensive only**, per Section 8.
4. **Cite OrigamiSpace as an arXiv preprint** unless a venue is confirmed; `papers.md` recorded a venue that its arXiv record does not carry.
5. **Every number carries a `fact : tag`** and `doccheck.mjs` must pass before submission. The tags survive in this file as LaTeX comments beside each number.
6. **Sections 9 to 11 stay empty until the runs exist.** If they cannot be filled in time, the paper is submitted as a benchmark paper and the ablation sentence leaves the abstract.

C DRAFT STATUS

Carried over from the header of `paper/DRAFT.md`. Sections 2 to 8 and 12 to 14 are written. Sections 9, 10 and 11 are empty and marked `[TO RUN]`, each saying which run would fill it. **Nothing goes into those three sections that has not been run.**

Ownership: `paper/DRAFT.md` owns the paper’s prose. It cites `DATASET.md`, `EXPERIMENTS_SETUP.md` and `notes/plan/*` rather than restating them. Numbers carry the repo’s inline `fact : tags`, so `doccheck.mjs` guards them there as everywhere else. Appendix B holds the editorial rules that are about writing rather than content.